

ADP470S	TOTAL MARK 100	
SUBJECT: ADP470S	Test 1	DUE DATE 8 JUNE 2023



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

COURSE:
BTech in Information Technology

SUBJECT COORDINATOR:

Mr T. Makhurane

LECTURERS / EXAMINERS:

Mr T. Makhurane

Question	Student Mark	Max. Score
1		20
2		20
3		20
4		20
		80

SPECIAL INSTRUCTIONS:

• Answer ALL 5 questions.

Hand-in question paper at the end of examination

REQUIREMENTS:

• Students should not leave within the first hour even if they have completed.

• Hand-in question paper together with the answer book at the end of examination

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QUESTION 1**[20 marks]**

- 1.1 Explain how the Factorial can be implemented using a recursive algorithm. [2]
 1.2 Explain how the Factorial can be implemented using an iterative algorithm. [4]
 1.3 Explain how the performance of the recursive algorithm for Factorial differs from the performance of the iterative algorithm? Hint: Both time and space complexity are important. [5]

Determine the Big-O expression for each of the following code excerpts as an expression involving n .

- 1.4

```
int sum = 0;
for (int i = 0; i < n; i++)
{
    sum = sum + a[i];
}
for (int j = 0; j < n; j++) {
    sum = sum + a[j];
}
```

 [3]
- 1.5

```
int prod = 1;
for (int i = 1; i < n; i++){
    for (int j = 1; j < n; j = j * 3){
        prod = prod * j;
    }
}
```

 [3]
- 1.6

```
int sum = 0;
for (i=0; i<=n-1; i+=3) {
    for (j=n-1; j>0; j--) {
        sum = sum + (i * j);
    }
}
```

 [3]

QUESTION 2**[20 marks]**

Given a queue based on an array implementation. If the queue stores an integer at each node, implement the following operations in an object-oriented programming language

- 2.1 Push. [5]
 2.2 Pop. [5]
 2.3 isEmpty. [5]
 2.4 Peek. [5]

QUESTION 3**[20 marks]**

- 3.1 Explain the working principles and time performance of the quick sort algorithm. Use this array as an example: {90, 23, 5, 109, 12, 22, 67, 34}. [12]
 3.2 Explain the time performance of the quick sort algorithm using asymptotic notation. [8]

QUESTION 4**[20 marks]**

4.1	Explain the working principles and time performance of the heap sort algorithm. Use this array as an example: {90, 23, 5, 109, 12, 22, 67, 34}}.	[12]
3.2	Explain the time performance of the heap sort algorithm using asymptotic notation.	[8]